

mm

5-1

61. What are the eigenvalues of the identity operator on R^n ? Describe each eigenspace.

$$Iv = 1v$$

$$\begin{array}{c} \downarrow \\ \lambda = 1 \\ \downarrow \\ R^n \end{array}$$

62. What are the eigenvalues of the zero operator on R^n ? Describe each eigenspace.

$$\underbrace{0}_{\text{matrix}} v = \underbrace{0}_{\substack{\text{I} \\ \text{scalar}}} v$$

$$\begin{array}{c} \downarrow \\ \lambda = 0 \\ \downarrow \\ R^n \end{array}$$

65. Suppose that 0 is an eigenvalue of a matrix A . Give another name for the eigenspace of A corresponding to 0 .

$$AV = 0v = 0$$

A 's null space

66. Prove that a square matrix A is invertible if and only if 0 is not an eigenvalue.

$$\text{invertible} \equiv \det A \neq 0$$

$$\det(A - \lambda I) \neq 0$$

$$\implies \det(A) \neq 0$$

$$\det(A - \lambda I) = 0$$

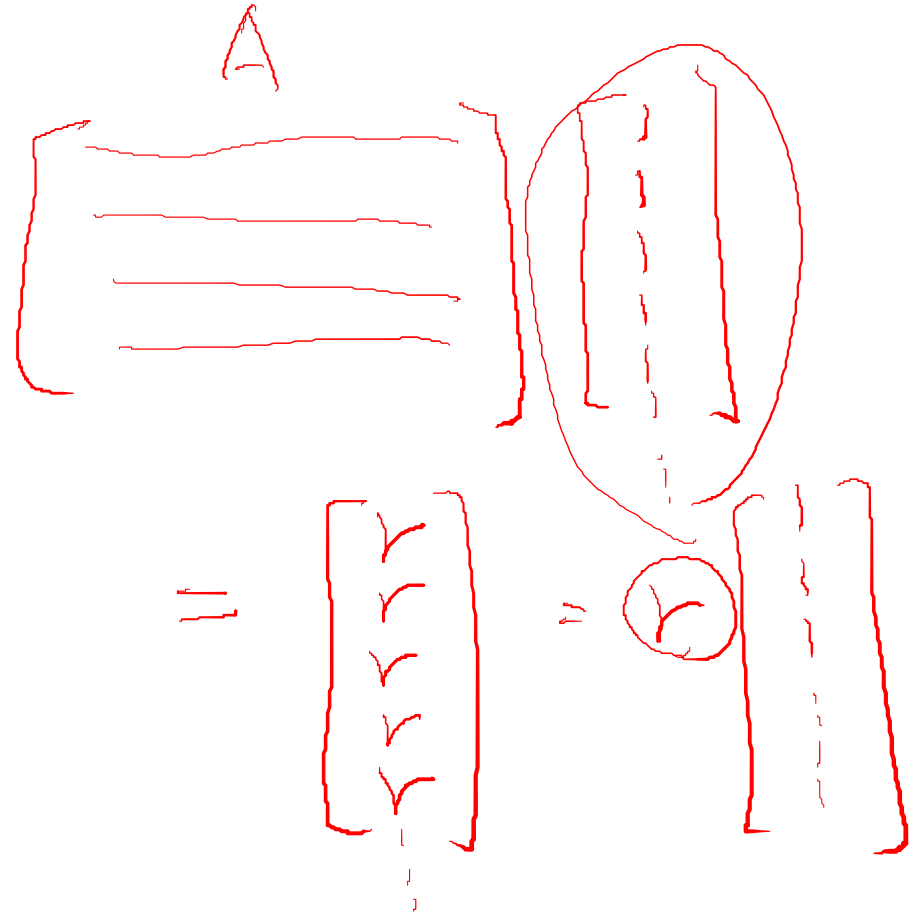
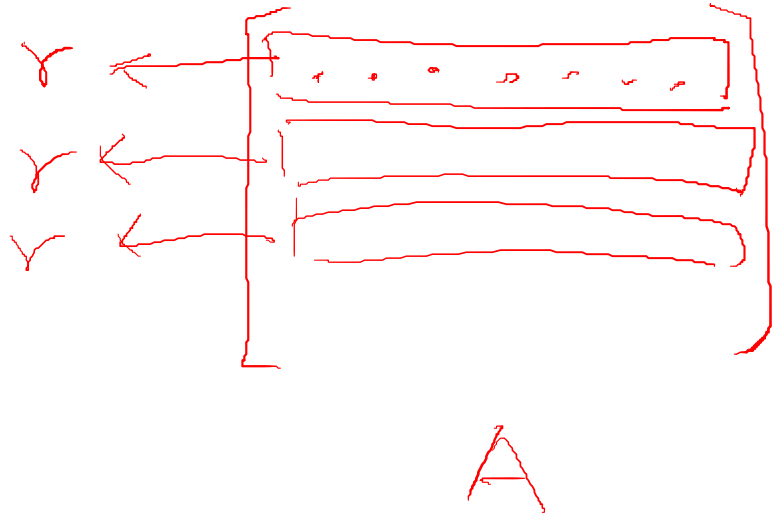
66. Prove that a square matrix is invertible if and only if 0 is not an eigenvalue.

67. Prove that if λ is an eigenvalue of an invertible matrix A , then $\lambda \neq 0$ and $1/\lambda$ is an eigenvalue of A^{-1} .

$$Av = \lambda v, \quad v \neq 0 \quad (\lambda \neq 0 \Leftrightarrow A \text{ is inv})$$

$$v = \lambda A^{-1}v \Rightarrow \frac{1}{\lambda}v = A^{-1}v$$
$$A^{-1}v = \boxed{\frac{1}{\lambda}}v$$
$$\lambda \neq 0$$

68. Suppose that A is a square matrix in which the sum of the entries of each row equals the same scalar r . Show that r is an eigenvalue of A by finding an eigenvector of A corresponding to r .



69. Prove that if λ is an eigenvalue of a matrix A , then λ^2 is an eigenvalue of A^2 .

$$Av = \lambda v$$

$$A^2v = A(\lambda v)$$

$$= \lambda^2 v$$

$$A \quad \lambda$$

$$A^n \quad \lambda^n$$

How about A^n ?

72. An $n \times n$ matrix A is called nilpotent if, for some positive integer k , $A^k = 0$, where 0 is the $n \times n$ zero matrix. Prove that 0 is the only eigenvalue of a nilpotent matrix. ✓

$$Av = \lambda v, \quad (v \neq 0)$$

$$\Rightarrow \underbrace{A^k}_{=0} v = A^{k-1}(\lambda v) = A^{k-2}(\lambda^2 v) = \dots = \underbrace{\lambda^k v}_{=0}$$

$$\Rightarrow \lambda^k = 0 \Rightarrow \lambda = 0$$

73. Let $\underline{v_1}$ and $\underline{v_2}$ be eigenvectors of a linear operator T on R^n , and let λ_1 and λ_2 , respectively, be the corresponding eigenvalues. Prove that if $\lambda_1 \neq \lambda_2$, then $\{v_1, v_2\}$ is linearly independent.

① $C_1 v_1 + C_2 v_2 = 0$ $\times \lambda_2$

$A(C_1 v_1 + C_2 v_2) = A0 = 0$

② $(C_1 \lambda_1) v_1 + (C_2 \lambda_2) v_2 = 0$

$(C_1 \lambda_2 - C_1 \lambda_1) v_1 = 0$

$\Rightarrow C_1 (\lambda_2 - \lambda_1) v_1 = 0$

$\Rightarrow C_1 = C_2 = 0$

74. Let $\underline{v_1}$, $\underline{v_2}$ and $\underline{v_3}$ be eigenvectors of a linear operator T on R^n , and let λ_1 , λ_2 and λ_3 , respectively, be the corresponding eigenvalues. Prove that if these eigenvalues are distinct, then $\{v_1, v_2, v_3\}$ is linearly independent.

Please refer to Theorem 5.3 (P317).